

KEYAN DING

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RESEARCH INTEREST

I am particularly interested in understanding how complex AI systems reason, generalize, and fail, and in developing evaluation and control methodologies that make AI systems more reliable, interpretable, and trustworthy. My long-term goal is to bridge my insights and experience from safety-critical engineering and modern AI research to build AI systems that can operate robustly in high-stakes environments.

EDUCATION BACKGROUND

University of Illinois at Urbana-Champaign

Aug 2021 - May 2023

M.S. in Mathematics

GPA: 3.5/4.0

- Courses Taken: Statistical Reinforcement Learning, Pattern Recognition, Online Learning, Theory of Probability, Mathematical Statistics

University of Washington, Seattle

Sep 2017 - Jun 2021

B.S. in Mathematics

GPA: 3.6/4.0

- Courses Taken: General Chemistry, Machine Learning, Data Structures and Algorithms, Mathematical Modeling, Linear Algebra, Topology, Stochastic Process

RESEARCH EXPERIENCE/TECHNICAL PROJECT

AI Design Review Assistant for Aviation Certification (site)

Apr 2026 - Now

Group Research Project | Python, RAG, LLM Evaluation

- **Built a retrieval-augmented generation (RAG)** assistant for FAA AC 21.17-4 (eVTOL Type Certification) that supports citation-grounded certification reasoning through structured document chunking and embedding-based retrieval.
- Ongoing Work: **Developing an aviation certification QA benchmark** for evaluating answer faithfulness, citation grounding, and uncertainty handling.
- Ongoing Work: **Investigating agentic design review workflows** that compare proposed system designs against certification guidance, identifying compliance gaps, unsupported assumptions, and missing safety evidence.

Enzyme Stability Prediction using Protein LLMs (code)

Dec 2025

Independent Research Project | Pytorch, ProtBERT Fine-tuning

- **Fine-tuned the ProtBERT transformer** (420M parameters) to predict enzyme melting temperature (T_m) from amino acid sequences, adapting NLP techniques to biological data.
- Optimized training efficiency on limited data by implementing **layer freezing strategies** and custom cosine learning rate schedules.
- Engineered a PyTorch training pipeline with **5-fold cross-validation** to mitigate overfitting and ensure generalization across protein sequences, utilizing model ensembling to maximize rank correlation.

Value Approximations in Offline Reinforcement Learning (pdf)

Aug 2022 - Dec 2022

Course Research Project | RL Theory

- Conducted a theoretical study of **offline reinforcement learning** algorithms, comparing FQI, MSBO, and SBEED through their concentrability assumptions, convergence guarantees, and horizon/sample complexities. Adapted each algorithm into the single-episode offline policy learning context by re-deriving every error bound and optimal policy function from scratch.

Geometric Optimization and Covering Problems (site)

Jan 2022 - Aug 2022

Graduate Mentor, Illinois Geometry Lab

- **Publication:** The PUMP Journal of Undergraduate Research, 2025
- **Co-hosted weekly research discussions** with undergraduate collaborators under Prof. Joseph Rosenblatt, translating open-ended geometric questions into tractable mathematical subproblems.
- Collaborated to address a **geometric optimization problem** to determine the maximum area covered by a fixed number of triangles in a disc.
- Employed rigorous mathematical techniques, including **constrained optimization (Lagrange Multipliers)** and **asymptotic analysis** to establish precise upper and lower bounds on the minimal error.

Selected Machine Learning Projects

- Built a hybrid recommendation model for the H&M Personalized Fashion Recommendations dataset, processing over 1M records and combining SVD-based latent factor modeling with temporal behavioral features(code).
- Developed a MATLAB image-classification pipeline using Fourier-transform features and gradient-based optimization, strengthening practical understanding of feature engineering and statistical pattern recognition (code).

PROFESSIONAL EXPERIENCE

System Engineer/Product Manager

Sep 2023 - Now

Commercial Aircraft Corporation of China, Ltd.

Top 1% in company performance in 2024

- **Led systems engineering work** for flight guidance functions in the Flight Management System (FMS) for COMAC narrow-body and wide-body aircraft programs.
- Translated high-level operational and safety objectives into **traceable system requirements, verification artifacts, and scenario-based test cases** for safety-critical flight guidance functions.
- **Developed an automated MATLAB/Simulink-based development workflow** that measures requirements coverage, executes regression test suites against new designs, and generates quantitative reports assessing functional and performance compliance to accelerate product iterations.
- Coordinated requirements across software, testing, autopilot, navigation, and pilot-facing stakeholders. Balanced competing operational, safety, and implementation constraints to **drive consensus on technically feasible system solutions**.
- Helped drive the development of a high-fidelity FMS prototype from early-stage architecture to approximately **95% functional coverage within the first year**, enabling rapid validation of core flight guidance concepts and system behaviors.

Organizer, Summer Illinois Math Camp (website)

Nov 2021 - May 2022

Department of Mathematics, UIUC

With research assistantship

- Organized a two-week math summer camp, overseeing budgeting, instructor/TA recruitment, and pedagogy training. Led weekly planning meetings, assigned tasks, and supervised execution.
- Managed communication with students and parents; designed promotional materials and conducted outreach across local schools.

PUBLICATION

Covering Area in a Disc by Finitely Many Triangles. (pdf)

Chen, Z., Ding, K., Huang, Y., Jia, X., Rosenblatt, J., Su, Y., & Xue, J. (2025).

The PUMP Journal of Undergraduate Research, 8, 47–56. <https://doi.org/10.46787/pump.v8i.3924>

TECHNICAL SKILL

Programming	Python, PyTorch, MATLAB/Simulink, C/C++, SQL, Java, Julia, R
ML	Transformer fine-tuning, cross-validation, reinforcement learning
AI Safety/Systems	LLM evaluation, retrieval-augmented generation, agent workflows, safety-critical systems engineering, requirements traceability, scenario-based testing, failure-mode analysis, certification-oriented documentation
Tools	Git, GitHub, Jupyter, LaTeX, Hugging Face, Linux command line
Speaking Languages	Mandarin(Native), English(Fluent)